Arista Access Node Diagnostic and Resolution SOP for QoS Congestion Event ALARM_8132 on MALL-ACC-21

Node: MALL-ACC-21 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8132

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a Critical QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Arista Access Node MALL-ACC-21 at the Malleswaram Node. The failure is attributed to the inherent design of the Arista EOS software, which manages traffic flows and QoS policies on the access node. Specifically, the issue arises from the interaction between the hardware buffers, the ingress policer, and the egress queuing mechanisms. When the ingress policer exceeds the buffer capacity, it can lead to microbursts and interface tail drops, causing QoS queue starvation. This scenario is often exacerbated by the high-speed nature of modern Ethernet interfaces and the dynamic traffic patterns present on the access node.

Pre-requisite Checks

- Verify that the Malleswaram Node is accessible remotely and physically.

- Check the status of the redundant link (if present) to ensure it's operational and not contributing to the congestion event.

- Confirm that all neighboring nodes and devices are functioning correctly and not generating excessive traffic.

Potential Root Causes

Cause	Probability
Hardware Buffer Overflow	High
Ingress Policer Exceeding Buffer Capacity	Medium
Egress Queue Starvation due to Microbursts	Low

Diagnostic Commands & Expected Outputs

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Command: show interface MALL-ACC-21 detail, Expected Output: Interface statistics, including errors and dropped packets., Description: This command isolates the interface-specific statistics, providing insight into the number of errors, dropped packets, and other relevant metrics that may indicate congestion.

Command: show qos policy MALL-ACC-21 detail, Expected Output: QoS policy configuration and statistics., Description: This command extracts the QoS policy configuration and statistics for the access
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node, enabling the identification of potential issues with policers, shapers, and other traffic management mechanisms.

Command: show buffer MALL-ACC-21 detail, Expected Output: Hardware buffer utilization and drop statistics., Description: This command provides detailed information on the hardware buffer utilization and drop statistics, helping to determine if the buffer overflow is contributing to the congestion event.
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Resolution Steps

Step 1: Step: Adjust Ingress Policer Settings, Command: config qos policy MALL-ACC-21 ingress-policer , Description: Adjust the ingress policer settings to reduce the rate at which traffic is policed, thereby preventing buffer overflows and reducing congestion.

Step 2: Step: Increase Egress Queue Sizes, Command: config qos policy MALL-ACC-21 egress-queue-size , Description: Increase the egress queue sizes to accommodate more packets, reducing the likelihood of queue starvation and congestion.

Step 3: Step: Implement Traffic Shaping, Command: config qos policy MALL-ACC-21 traffic-shaper , Description: Implement traffic shaping to regulate the rate at which traffic is transmitted, preventing microbursts and reducing congestion.

Post-Check Verification

```
Command: show interface MALL-ACC-21 detail, Expected Output: Improved interface statistics, with reduced errors and dropped packets., Description: Verify that the interface statistics have improved, indicating that the resolution steps have been effective in reducing congestion.

Command: show qos policy MALL-ACC-21 detail, Expected Output: Improved QoS policy configuration and statistics., Description: Verify that the QoS policy configuration and statistics have improved, indicating that the resolution steps have been effective in managing traffic.

Command: show buffer MALL-ACC-21 detail, Expected Output: Improved hardware buffer utilization and drop statistics., Description: Verify that the hardware buffer utilization and drop statistics have improved, indicating that the resolution steps have been effective in reducing congestion.
```

Roll-back Procedures

- Step: Restore Original Ingress Policer Settings, Command: config qos policy MALL-ACC-21 ingress-policer , Description: Restore the original ingress policer settings to revert to the previous configuration.

- Step: Restore Original Egress Queue Sizes, Command: config qos policy MALL-ACC-21 egress-queue-size , Description: Restore the original egress queue sizes to revert to the previous configuration.

- Step: Restore Original Traffic Shaping Configuration, Command: config qos policy MALL-ACC-21 traffic-shaper , Description: Restore the original traffic shaping configuration to revert to the previous setup.